

Unit IV: Probability, Statistics, and Mathematical Tools

Study notes for AI, probabilistic reasoning, and quantum measurement intuition

Learning Goals

- Understand probability as a language for uncertainty.
- Describe random variables and compute expectation in simple cases.
- Use mean, variance, and standard deviation to summarize data and uncertainty.
- Recognize common probability distributions conceptually.
- Connect complex numbers and inner products to vectors, signals, and quantum states.

1. Basics of Probability

Probability measures how likely an event is. A probability is always between 0 and 1, where 0 means impossible and 1 means certain. In AI, probability helps a system act sensibly when information is incomplete, noisy, or changing.

$$0 \leq P(A) \leq 1$$

$$P(\text{sample space}) = 1$$

$$P(\text{not } A) = 1 - P(A)$$

Key terms

- Experiment: a process with uncertain outcome, such as tossing a coin or observing sensor data.
- Sample space: the set of all possible outcomes.
- Event: a subset of the sample space, such as rolling an even number.
- Independent events: events where knowing one occurred does not change the probability of the other.
- Conditional probability: the probability of A when B is known to be true.

$$P(A \mid B) = P(A \text{ and } B) / P(B), \text{ when } P(B) > 0$$

Example: If a classifier says an email is spam, conditional probability helps estimate the chance that the email truly is spam given the classifier's evidence.

2. Random Variables and Expectation

A random variable assigns a numerical value to each outcome of a random experiment. It can be discrete, such as the number of heads in three coin tosses, or continuous, such as temperature or waiting time.

$$\text{Expectation: } E[X] = \sum x \cdot P(X = x)$$

$$\text{Continuous idea: } E[X] = \int x \cdot f(x) \, dx$$

Expectation is the long-run average value of a random variable. It is not always the value you will actually observe; it is the balance point of the distribution.

- If a game pays 10 points with probability 0.3 and 0 points with probability 0.7, then $E[X] = 10(0.3) + 0(0.7) = 3$ points.
- In AI, expected value is used to compare actions under uncertainty, such as choosing the route with the lowest expected travel time.

3. Mean, Variance, and Standard Deviation

The mean describes the center of data or a distribution. Variance describes how spread out the values are. Standard deviation is the square root of variance, so it uses the same units as the original data.

Mean: $\mu = E[X]$

Variance: $\text{Var}(X) = E[(X - \mu)^2]$

Standard deviation: σ

Measure	Meaning
Mean	Typical or average value.
Variance	Average squared distance from the mean; larger means more spread.
Standard deviation	Typical distance from the mean, measured in original units.

Low variance means predictions are tightly clustered. High variance means there is more uncertainty or instability. In machine learning, variance also describes how much a model's predictions change when trained on different data.

4. Common Probability Distributions

A probability distribution describes how probability is assigned across possible values. You do not always need the full formula first; the most important starting point is knowing what kind of situation each distribution models.

Distribution	Conceptual use
Bernoulli	One yes/no outcome, such as success/failure or true/false.
Binomial	Number of successes in a fixed number of independent trials.
Uniform	All outcomes in a range are equally likely.
Normal	Bell-shaped variation around a mean; common for measurement noise.
Poisson	Counts of events in a fixed interval, such as arrivals or defects.
Exponential	Waiting time until the next event in a Poisson-like process.

Choosing a distribution is a modeling decision. For example, image sensor noise may be approximated with a normal distribution, while the number of requests arriving at a server in one minute may be modeled with a Poisson distribution.

5. Complex Numbers

A complex number has a real part and an imaginary part. It is written as $a + bi$, where i is defined by $i^2 = -1$. Complex numbers are useful for representing rotations, waves, signals, and quantum amplitudes.

$$z = a + bi \quad \text{Re}(z) = a \quad \text{Im}(z) = b \quad |z| = \sqrt{a^2 + b^2}$$

- Addition: add real parts and imaginary parts separately.
- Multiplication: distribute terms and use $i^2 = -1$.
- Magnitude: distance from the origin in the complex plane.
- Complex conjugate: if $z = a + bi$, then $\text{conjugate}(z) = a - bi$.

In quantum theory, complex numbers can encode probability amplitudes. The observable probability is related to the squared magnitude of an amplitude, not the amplitude alone.

6. Inner Products

An inner product is a way to multiply two vectors and get a scalar. It measures alignment or similarity. In ordinary real vectors, it is the dot product. In complex vector spaces, the conjugate of one vector is used.

$$\text{Real vectors: } \langle x, y \rangle = x_1 y_1 + x_2 y_2 + \dots + x_n y_n$$

Complex vectors: $\langle x, y \rangle = \text{conjugate}(x_1) \cdot y_1 + \dots + \text{conjugate}(x_n) \cdot y_n$

- If the inner product is large and positive, the vectors point in similar directions.
- If the inner product is zero, the vectors are orthogonal, meaning they carry no shared direction under this measure.
- In machine learning, inner products appear in similarity search, neural networks, kernels, and embeddings.
- In quantum mechanics, inner products help calculate measurement probabilities between states.

Applications

Uncertainty in AI

AI systems often act without complete certainty. A speech recognizer may be unsure between two words, a medical model may estimate disease risk, and a robot may receive noisy sensor readings. Probability gives these systems a disciplined way to represent uncertainty instead of pretending every decision is certain.

Probabilistic Models

Probabilistic models describe relationships between variables using probability. Examples include Naive Bayes classifiers, Bayesian networks, hidden Markov models, and probabilistic graphical models. They are useful when evidence is incomplete and the system must update beliefs as new information arrives.

Quantum Measurement Intuition

Quantum states can be represented using vectors with complex-number amplitudes. Measurement does not simply reveal a pre-written classical value; it gives outcomes according to probabilities. A simplified intuition is: amplitudes describe the state, inner products compare possible states, and squared magnitudes give measurement probabilities.

Quick Revision Sheet

- Probability is a number from 0 to 1 that measures uncertainty.
- Conditional probability updates likelihood when evidence is known.
- A random variable turns uncertain outcomes into numbers.
- Expectation is the long-run average or balance point.
- Variance and standard deviation measure spread.
- Distributions are reusable patterns for uncertainty.
- Complex numbers represent two-dimensional quantities and are essential for waves and quantum amplitudes.
- Inner products measure similarity, projection, and quantum measurement overlap.

Practice Questions

- A coin is tossed three times. What is the sample space? Define a random variable for the number of heads.
- A model predicts rain with probability 0.7. What does this mean in practical decision-making?
- Why can two datasets have the same mean but different standard deviations?

- Choose a suitable distribution for counting website visits per minute. Explain your choice.
- Compute the magnitude of $z = 3 + 4i$.
- What does it mean for two vectors to have inner product zero?
- How does probability help an AI system work with noisy data?
- Why are complex numbers useful in quantum measurement intuition?